

Aditi Roy Adri

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

OBJECTIVE

Motivated CSE student with a strong foundation in algorithms, machine learning, systems programming, and full-stack development. Seeking opportunities to apply technical expertise, contribute to impactful projects, and grow in a collaborative environment.

EDUCATION

BRAC University, Dhaka

2023 — 2026

Bachelor of Science in Computer Science and Engineering

CGPA: 3.84

Holy Cross College, Dhaka

2021

Higher Secondary Certificate (HSC)

GPA: 5.00

Narayanganj High School and College, Narayanganj

2019

Secondary School Certificate (SSC)

GPA: 5.00

TECHNICAL SKILLS

Programming Languages

Python, C++, C, Java, PHP, Assembly (Emu8086 / x86), Bash

Machine Learning & Computer Vision

Deep Learning, CNNs, Image Segmentation, Object Detection, Transfer Learning, Model Evaluation, YOLO, U-Net

Web Technologies

HTML5, CSS3, JavaScript, React.js

Libraries & Frameworks

PyTorch, NumPy, Pandas, Scikit-Learn, Matplotlib, Jupyter Notebook

Databases

MySQL, PostgreSQL

Graphics & Networking

OpenGL (PyOpenGL), Cisco Packet Tracer

Tools & Platforms

Git, GitHub, LaTeX, Linux, VS Code

Soft Skills

Communication, Critical Thinking, Problem-Solving, Teamwork

PROJECTS

CARDD — Car Damage Detection with YOLO [GitHub](#)

Jun 2026

Python · YOLO11 · YOLOv8 · PyTorch · Object Detection · Computer Vision

- Trained and benchmarked YOLO11-L against YOLOv8-L on the CarDD dataset — 4,000 images, 8,740 annotated boxes across 6 damage classes (scratches, dents, cracks, broken lamps, shattered glass, flat tires). YOLO11-L achieved **mAP@50-95 of 0.557**.

Pet Image Segmentation & Breed Classification (U-Net) [GitHub](#)

May 2026

Python · PyTorch · Jupyter Notebook · ResNet-34 · Oxford-IIIT Pet Dataset

- Built U-Net and Attention U-Net architectures for binary pet image segmentation and 37-breed classification. Achieved **~88% IoU** using a pretrained ResNet-34 encoder; from-scratch models reached ~80% IoU. Implemented data augmentation, combined segmentation + classification loss, and ablation studies.

SaveLife — SOS Emergency Response App [GitHub](#)

Jul 2026

React · Vite · Node.js · Express · PostgreSQL · JWT

- Emergency response platform with one-tap SOS alerts and GPS geolocation, a live responder dashboard, and role-based JWT authentication. React/Vite frontend, Node.js/Express API, PostgreSQL, deployed on Render.

Roaming Sonic — Travel & Tour Management Platform [GitHub](#) [Live](#)

Jan 2026

MongoDB · Express.js · React.js · Node.js · Vercel

- Full-stack MERN web platform for Bangladesh tourism featuring tour packages, hotel bookings, bus ticketing, guide services, AI chatbot, and referral rewards system. Live-deployed on Vercel.

Metadata Journal System [GitHub](#)

Jan 2026

C · Operating Systems

- OS-level implementation of file metadata management and journaling mechanisms to maintain filesystem consistency across crashes and concurrent access.

EXPERIENCE & ACTIVITIES

- **Women & Child Affairs Secretary**, Executive Committee — Bangladesh Youngster Social Organization (BYSO), Dhaka District Unit, for the 2026–27 session. Leadership role supporting women's and children's welfare initiatives.
- Competitive programming on Codeforces — 500+ problems solved in Python across multiple difficulty levels.
- Participation in university-level and inter-university programming contests at BRAC University.
- Active volunteer in community and social-welfare activities.
- Vocalist for cultural clubs and events at college and university.

INTERESTS

Machine learning, image processing, computer graphics, competitive programming, full-stack web development, movies and TV series.